Diego **Zamboni**

Organizational Leader · Senior Security Architect · Computer Scientist

diego@zzamboni.org · zzamboni.org · zzamboni · zzamboni · zzamboni

I am a Senior Organizational Leader, Enterprise Security Architect and Computer Scientist with 32 years of experience building systems and leading teams and projects. I specialize in the areas of Computer Security, Cloud Computing and Configuration Management. I possess a strong combination of leadership, conceptual and technical knowledge with excellent communication skills.

Key skills

Leadership	32 years of team and project leadership experience; IT Enterprise Architecture.
Computer Security	Enterprise security architecture; virtualization and cloud computing security; governance, risk management and compliance; intrusion detection and prevention; software security and secure software development.
Cloud Computing	AWS architecture, infrastructure, development and security.
Communications and Research	Excellent written and spoken communication skills; research and teaching experience.
Research	Ph.D. in Computer Science, 9 years of experience at IBM Research.
Languages	Spanish (native), English (100%), German (B2 level).

Experience

Amazon Web Services

2021 – Present *Switzerland*

Senior Global Security Architect *Oct 2021 – Present*

- I work with customers and within AWS to increase security awareness, design and build secure solutions, mentor and develop colleagues and customers in security best practices.
- As Lead Security Architect at the Stellantis *Virtual Engineering Workbench* (VEW) project, I have:
  - Established the VEW security workstream, define and promote security best practices and drive activities related to risk analysis, threat modeling and mitigation.
  - Established measurement mechanisms for status and metrics for security activities, which provide visibility to both technical and business stakeholders.
  - Established a *Security Champions* program to promote and transfer security knowledge.
  - Define and track security features to fulfill customer business requirements and AWS best practices.

Global Security Architect *Sep 2021 – Sep 2022*

- Worked with AWS global customers to improve security posture and promote secure design and implementation practices.
- In the Volkswagen Digital Production Platform (DPP) program, I established scalable and durable mechanisms to enable DPP partners to work securely in the DPP program. I also created and promoted security learning materials tailored for various roles within the project, and increased security awareness and knowledge by promoting a *Security Guardians* initiative across the organization.

Swisscom

2014 – 2021 *Switzerland/U.S.A.*

Enterprise Architect and IT Clouds Solution Security Architect *Apr 2019 – Sep 2021*

- As a Solution Security Architect for IT Clouds, I was a member of the *Swisscom IT Clouds* leadership team:
  - I defined and manage yearly KPIs and the security features and characteristics of Swisscom's cloud offerings.
  - I ensured cloud platform and service compliance with internal, contractual and regulatory standards (e.g. ISO27001, ISAE3402/3000, FINMA and GDPR).
  - I established and manage the *Security Champions* initiative to promote and improve a culture of security responsibility.
  - I established organization-wide processes for risk management, threat modeling, audits and penetration testing.
  - I was the point of contact for CISO-level interactions with Swisscom's cloud customers.
- As an Enterprise Architect, I defined future Swisscom products and solutions as part of the *Swisscom Enterprise Architecture* team.

Architect and Engineering Leader for Monitoring, Health and State Management — *Aug 2014 – Apr 2019*

- Starting as a project lead and architect, I founded and led a team of up to 16 people that designed, implemented and operated the monitoring and logging capabilities for Swisscom's IaaS and PaaS evolving cloud offerings:
  - Unified diverse monitoring components into a cohesive framework, reducing costs and increasing efficiency by 20%.
  - Defined architecture and implemented customer-facing monitoring and logging services.
  - Designed and implemented the *Orchard* monitoring platform for Swisscom's *Application Cloud*.
  - Defined and implemented integrations between Jira and OpsGenie for alerting of user-reported incidents.
  - Managed business relationship and implemented Swisscom-wide use of OpsGenie for alert management.
  - Managed engineering relationship with VMware for integration of their products in Swisscom's monitoring systems.

CFEngine AS — *2011 – 2014* — *Norway/U.S.A. (remote)*

Senior Security Advisor and Product Manager — *Oct 2011 – Jun 2014*

- Developed the CFEngine language roadmap and security strategy.
- Acted as overall CFEngine Advocate, with a special focus on security.
- Published the book Learning CFEngine (O'Reilly Media), which was used for promotion and education.
- Established and led the *CFEngine Design Center* project through its initial implementation phase.

HP Enterprise Services — *2009 – 2011* — *Mexico*

Account Security Officer and Service Delivery Consultant — *Nov 2009 – Oct 2011*

- Managed security-related topics for HP enterprise customers in Mexico and abroad.
- Managed all security-related aspects of the design, implementation and delivery of IT Outsourcing projects.
- Analyzed, designed and implemented solutions in the areas of system automation, configuration management, system administration, system design, virtualization, performance and security.

IBM Zurich Research Lab — *2001 – 2009* — *Switzerland*

Research Staff Member — *Oct 2001 – Oct 2009*

- Established and developed research projects in intrusion detection, malware containment, and virtualization security, including:
  - *Project Phantom*: Security for VMware virtual environments using virtual machine introspection.
  - *Billy Goat*: An active worm-detection and capture system, deployed in the IBM internal and external networks.
  - *Router-based Billy Goat*: An active worm-capture device deployed at the network boundary and coupled with a border router to effectively and automatically spoof every unused IP address outside the local network.

National Autonomous University of Mexico (UNAM) — *1991 – 1996* — *Mexico*

Head of Computer Security Area and System Administrator — *Nov 1991 – Aug 1996*

- Established UNAM's first Intrusion Response Team, which evolved into the current Information Security organization.
- Managed security services for the supercomputing organization and for the whole University, including incident response, security information, auditing and teaching.
- Established the *International Computer Security Day*, which has grown and divided into the *Computer Security Day* (a one-day event) and the *Seguridad en Cómputo* (Computer Security) conference (a multi-day event).
- System Administrator at UNAM's Supercomputing Center, managing UNAM's Cray Y-MP Supercomputer and related systems.

Education

Ph.D. in Computer Science, Purdue University — *West Lafayette, IN, U.S.A.* — *1996–2001*

Using Internal Sensors for Computer Intrusion Detection, Advisor: Eugene H. Spafford.

Certifications



Research, Publications, Teaching and References

Books: Learning CFEngine (2012), Learning Hammerspoon (2018), Literate Configuration (2019), Publishing with Emacs, Org-mode and Leanpub (2021).

References available upon request. Full Curriculum Vitæ available at https://zzamboni.org/vita/.